Myofascial Release (MFR) Summary

1) Pathological Process

Appearance: Occurs when a body segment stops receiving appropriate stimuli, establishing a pathological process characterized by deficient circulation and limitation in nutrient supply to the fundamental substance of connective tissue, leading to densification.

Result: Dense tissue becomes hypomobile, leading to movement limitations.

Myofascial Pain

Origin: Both muscle and fascia.

Symptoms: Deep, sharp, burning, dull, diffuse, heavy, and squeezing pain.

> ***Important Note:*** *Needs to be treated quickly.*

Myofascial Therapy

Goal: Therapy which can directly change and improve the health of the fascia.

Purpose of MFR:

- Break down scar tissue.
- Relax the muscle and myofascial tissues.
- Restore good posture.
- Focus on relaxing deep tissues of the body.
- Provide lasting and effective relief to patients.

Fixes / Anchors

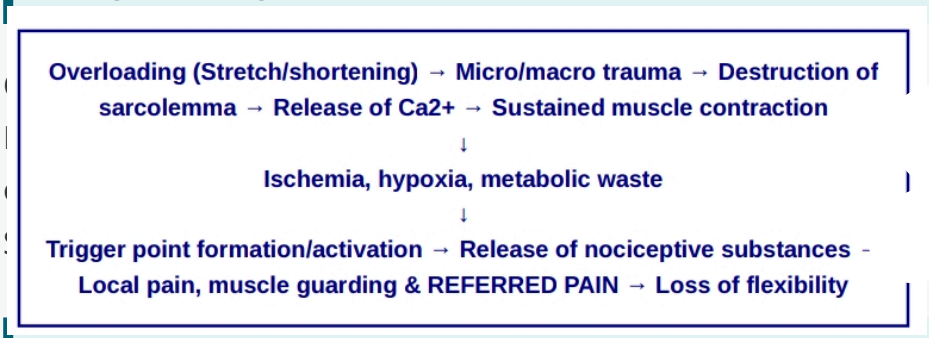
Forearm, Elbow, Fist, Palm, Grip, Reinforced thumbs, Finger(s), Knuckle(s), External devices.

MFR Depends On

Direction of Stretch: Horizontal, Diagonal, Perpendicular, Vertical.

Other Factors: Force, Duration.

Why Do Myofascial Release Instead of Gross Stretch?



MFR Definition

MFR therapy is a combination of manual traction and prolonged assisted stretching maneuvers designed to break up fascial adhesions.

Effects of MFR

Mechanical Effects

- Reduce muscle tension.
- Increase circulation.
- Pain relief.
- Increase mobility.

Neurological Effects

- Inducing segmental reflexes.
- Inducing suprasegmental reflexes.

Pain Gate Theory

Stimulation of mechanoreceptors by MFR may close the "Pain Gate" through presynaptic and postsynaptic inhibition. Additionally, it induces the release of endogenous opiates.

Fibromyalgia: MFR was found effective in reducing muscle pain in patients with Fibromyalgia (FM).

Why MFR is Effective?

MFR more directly targets proposed peripheral pain generators residing in the fascia. Benefits were evaluated on pain, disease severity, and patients with fibromyalgia.

Neurobiological Explanation

Schleip Theory: Researchers suggested that change in tone is not achieved by altering fascia shape, but is instead controlled by the nervous system. Sustained manual pressure may work through **Ruffini Corpuscles**.

Why Ruffini Corpuscles?

Clinical Observation: Applying a slow, extended stretch to the skin can create local effects (decreased tension in the treated area) and central effects (overall sense of relaxation).

Ruffini Corpuscles: Characteristics

- Respond to lateral skin stretch or stretching the skin along the same plane as underlying tissue.
- **Slow-adapting receptors:** They continue firing as long as stretch is maintained.

Mechanoreceptor Stimulation

Hands on skin → Mechanoreceptors stimulated → Impulses travel through sensory nerves → Reach the brain → Brain evaluates/responds → Sends impulses through nerves → Changes in blood vessel diameter, breathing, and muscle tone.

Brain Response to Touch

- **If the Brain Likes the Touch:** Relaxation, release of tension, decreased pain sensation.
- **If the Brain Feels Threatened:** Increased tension, increased protective responses.

Benefits of MFR

Less expensive, patient ease, non-invasive, self-treatment, and effective release of tightness/restrictions.

General Indications

Myofascial restrictions, pain not alleviated by other treatments, complex/global/specific pain, underlying chronic conditions causing tightness, complex postural asymmetries, and athletes needing stretching to increase performance or prevent injury.

Precautions / Contraindications

Avoid or use with caution in: Inflammation, patient uncomfortable with touch, dermatitis, contagious/infectious diseases, circulatory problems, fractures, open wounds, taking blood clotting medicine, and osteoporosis.

MFR Treatment Steps

Step	Description
Step 1	Gross stretch of entire body area.
Step 2	Focused stretch of muscles in the same body area.
Step 3	Finish with gross stretch of the entire muscle.

Focused Stretch

Technique: One hand, finger, or body weight acts as an anchor. Use a broad surface for large muscles; for smaller muscles, use 1–2 fingers.

Hand Placement: 1–2 fingers of each hand; one or both hands may move. Distance between fingers may be several millimeters or several inches.

Levels of MFR

- **Level 1:** No Stretch / Pre-tension.
- **Level 2:** Pre-tension (Stripping).
- **Level 3:** Moderate Passive Stretch.
- **Level 4:** Active MFR.

Common Mistakes

Too much pressure, too long of stretch, too many applications.